## General Questions

1. (a) Describe the functionality of the following command:

```
sh < proc > dat
```

(b) How are terminal processes put in the background? Which command shows these backgrounded processes?

(c) How many positional parameters does a shell script accept by default (without any extra help)?

(d) Which special shell variable represents the number of arguments passed on the command-line to the script?

## Shell scripts

2. The following shell script contains **3** syntax errors. Find the errors by providing the line they appear in. Write the corrected line beside it.

```
1  #!/bin/sh
2  $counter=1
3  number1=1
4  number2=$number1
5  echo $number1
6  echo $number2
7  while [ counter -lt 20 ]; do
8      result=`expr $number1 + $number2`
9      echo $result
10     number1=$number2
11     number2=$result
12     counter=`expr $counter++`
13 done
```

Enter the line number here and beside it the corrected version.

Line: _____ Correction: _____

Line: _____ Correction: _____

Line: _____ Correction: _____

What is the output of the program, after all the errors were fixed?

## Permissions in UNIX filesystems

3. In the following table, provide the changes that are set **after** executing the `chmod` command (middle column) on the original permissions (left column).

| No. | Old permission | `chmod` | New permissions |
|---|---|---|---|
| 1 | `r-x--xrw-` | 751 | _____ |
| 2 | `r---w-rw-` | 425 | _____ |
| 3 | `rwxr-xr--` | 632 | _____ |
| 4 | `r-xr-----` | 547 | _____ |
| 5 | `rw-rw-r--` | 324 | _____ |

Provide the calculations for each permission set below.

## Multiple Choice Questions

4. (a) Given the file `homepage.txt` with the following line:

```
https://www.fbi.h-da.de/personen/reuschling%20benedict/unix%20for%20software%
20developers.html
```

Change this by replacing **all** `%20` with (`-`). Find and check the proper `sed`-line that does this.
**Hint:** *One* solution is correct.

- ☐ `cat homepage.txt | sed s/\%20/-`
- ☐ `cat homepage.txt | sed s/%20/-`
- ☐ `cat homepage.txt | sed s/%20/-/`
- ☐ `cat homepage.txt | sed s/%20/-/g`

(b) Which of the following instructions throws an error? **Hint:** there is *one* correct solution.

- ☐ `result=$(expr 12 + 3)`
- ☐ `result=$(expr 12 * 3)`
- ☐ `result=$(expr 1 - 3)`
- ☐ `result=$(expr 12 / 3)`

(c) Which global variable stores the exit code of the previous command?
**Hint:** there is *one* correct solution.

- ☐ `$0`
- ☐ `$1`
- ☐ `$?`
- ☐ `$$`

(d) Given the following files:

```
?.t?t       ?.txt       a.txt       abc.txt       test.txt
```

Check all proper solutions that delete the file `?.t?t`. Other files with the extension `.txt` must **not** be deleted!
**Hint:** there is *more than one* correct solution.

- ☐ `rm \?.t\?t`
- ☐ `rm ?.t\?t`
- ☐ `rm "?.t?t"`
- ☐ `rm *.t*t`
- ☐ `rm '?.t?t'`

(e) What is the terminating character in a shell `switch` statement between `case` patterns?
**Hint:** there is *one* correct solution.

- ☐ `;;`
- ☐ `!$`
- ☐ `end`
- ☐ `#!`

## Understanding Shell Scripts

5. Look at the following script. Sketch the layout of the filesystem **after** the script executed (called with `root` permissions). Provide the permissions and ownership for each directory created.

```
#!/bin/sh
homedir="springfield"
i=2
mode=$i
mkdir /$homedir
for n in maggie bart lisa marge homer; do
    if [ `expr $i % 2` -eq 0 ]; then
        i=`expr $i + 1`
        mode=$i
    else
        i=`expr $i \* $i`
        mode=`expr $i - 1`
    fi
    if [ $i -gt 7 ]; then
        i=4
        mode=$i
    else
        i=`expr $i + 1`
    fi
    mkdir /$homedir/$n
    chmod ${mode}${mode}${mode} /$homedir/$n
    chown $n /$homedir/$n
done
```